

Engineering Reboot 2026

Week-02 Field Manual v2.0 (LOCKED)

Governance Verification

New Directories Introduced: None

New Artifact Types Introduced: None

Evidence.md Required: No

Day 06

Topic: Conditionals

Exercise File: Exercises\Day-06\decision_program.py

Required Artifacts

Notes file, Journal file, Exercise file, Evidence screenshots, Git commit

Program Requirements

- Input: integer age
- Valid range: 0-120
- Output categories: Child(0-12), Teenager(13-17), Adult(18-64), Senior(65-120)
- Invalid ages: <0 or >120

Required Test Cases

- 5->Child
- 16->Teenager
- 30->Adult
- 70->Senior
- -1->Invalid age
- 150->Invalid age

Evidence Screenshots

screenshot-01-git-status.png

screenshot-02-code.png

screenshot-03-program-output.png

screenshot-04-git-commit.png

Git Commands

git status

git add .

git commit -m "Complete Week 02 Day 06"

End-of-Day Success Criteria

Exercise completed, notes updated, journal updated, screenshots captured, commit completed.

Day 07

Topic: Boolean Logic

Exercise File: Exercises\Day-07\eligibility_checker.py

Required Artifacts

Notes file, Journal file, Exercise file, Evidence screenshots, Git commit

Program Requirements

- Inputs: age and driver's license status
- Use and/or/not operators
- Determine eligibility

Required Test Cases

- 18 + license=True -> Eligible
- 17 + license=True -> Not Eligible
- 18 + license=False -> Not Eligible

Evidence Screenshots

screenshot-01-git-status.png

screenshot-02-code.png

screenshot-03-program-output.png

screenshot-04-git-commit.png

Git Commands

git status

git add .

git commit -m "Complete Week 02 Day 07"

End-of-Day Success Criteria

Exercise completed, notes updated, journal updated, screenshots captured, commit completed.

Day 08

Topic: Loops

Exercise File: Exercises\Day-08\counting_program.py

Required Artifacts

Notes file, Journal file, Exercise file, Evidence screenshots, Git commit

Program Requirements

- Display 1-10
- Display 10-1
- Display even numbers 2-20
- Display odd numbers 1-19

Required Test Cases

- Verify all sequences print correctly

Evidence Screenshots

screenshot-01-git-status.png

screenshot-02-code.png

screenshot-03-program-output.png

screenshot-04-git-commit.png

Git Commands

git status

git add .

git commit -m "Complete Week 02 Day 08"

End-of-Day Success Criteria

Exercise completed, notes updated, journal updated, screenshots captured, commit completed.

Day 09

Topic: Loop Practice

Exercise File: Exercises\Day-09\pattern_generator.py

Required Artifacts

Notes file, Journal file, Exercise file, Evidence screenshots, Git commit

Program Requirements

- Generate star pyramid pattern
- Generate numeric growth pattern
- Use nested loops

Required Test Cases

- * * * * *
- 1 12 123 1234 12345

Evidence Screenshots

screenshot-01-git-status.png
screenshot-02-code.png
screenshot-03-program-output.png
screenshot-04-git-commit.png

Git Commands

git status
git add .
git commit -m "Complete Week 02 Day 09"

End-of-Day Success Criteria

Exercise completed, notes updated, journal updated, screenshots captured, commit completed.

Day 10

Topic: Mini Project

Exercise File: Exercises\Day-10\menu_driven_calculator.py

Required Artifacts

Notes file, Journal file, Exercise file, Evidence screenshots, Git commit

Program Requirements

- Menu options 1-5
- Support add/subtract/multiply/divide
- Accept integer and decimal input
- Prevent divide-by-zero
- Loop until Exit selected

Required Test Cases

- $10+5=15$
- $10-5=5$
- $10*5=50$
- $10/5=2$
- $10/0 \rightarrow$ Cannot divide by zero
- 99 $\rightarrow$ Invalid menu option

Evidence Screenshots

screenshot-01-git-status.png

screenshot-02-code.png

screenshot-03-program-output.png

screenshot-04-git-commit.png

Git Commands

git status

git add .

git commit -m "Complete Week 02 Day 10"

End-of-Day Success Criteria

Exercise completed, notes updated, journal updated, screenshots captured, commit completed.